MATTIA PAGETTI

Senior Data Engineer

@ pagetti.mattia@gmail.com in mattia-pagetti-40891019b Born in Sondrio (SO), Italy, 21/02/1994

github.com/Paggio

WORK EXPERIENCE

Senior Data Engineer ← Data Engineer

TrueLayer

02/2022 - 01/2026 London, UK / Milano, IT

- *I developed and managed data pipelines aggregating information from microservices databases, queues, third-party APIs and Elastic Search, organizing it in S3 (Parquet/Iceberg format). Data is processed using **Spark** on **EMR** and queried via Athena, orchestrated through **Airflow** DAGs or **Kubernetes** microservices.*
- *As a Senior Data Engineer (a role to which I was promoted in February 2023), I designed and implemented a new data platform from the ground up, automating Aurora cluster snapshots and their load into **Redshift** through **ECS** tasks, **Glue**, and Step Functions, with data models managed via **DBT**. All infrastructure provisioned using **Terraform**.*

Big Data Engineering Junior Specialist

Illimity

05/2020 - 02/2022 Milano, IT

- *I contributed to the development and maintenance of Illimity's data streams within an Azure-based IT architecture. The data streams were built using **Databricks**, **Data Factory**, **Delta Lake**, and **MongoDB**, with all workflows managed through Azure DevOps*

Doctoral Student

Max Planck Institute for the Structure and Dynamics of Matter

09/2019 - 04/2020 Hamburg, Germany

- *I participated in the theoretical and computational analysis of RIXS cross-sections, utilizing software for conducting advanced simulations of realistic condensed matter systems.*

KEY SKILLS

Main Stack

- *Spark, Airflow, Iceberg, DBT, Databricks, Terraform, Kubernetes, LangChain/LangGraph*

AWS Tools

- *EMR, Redshift, Athena, S3, Glue, EC2, ECS, Lambda, Step Functions, Redshift, RabbitMQ, Step Functions, EKS*

Microsoft Azure Tools

- *Data Factory, DataLake, Purview, Synapse, MongoDB*

EDUCATION

Master's Degree in Physics Engineering

Politecnico di Milano

02/2017 - 04/2019 Milano (MI), IT

- *Nanophysics and Nanotechnology*
- *Final Grade: 108/110*

Bachelor's Degree in Physics Engineering

Politecnico di Milano

10/2013 - 02/2017 Milano (MI), IT

- *Final Grade: 91/110*

PUBLICATIONS

Determining the Electron-Phonon Coupling in Superconducting Cuprates by Resonant Inelastic X-ray Scattering: Methods and Results on $Nd_{1+x}Ba_{2x}Cu_3O_{7\delta}$

Politecnico di Milano

2017 https://arxiv.org/abs/1906.01270

- *In high critical temperature superconductors the actual role of electron-phonon coupling (EPC) investigated.*